1 It is given that $y = \sinh(x^2) + \cosh(x^2)$.

(a) Use standard results from the list of formulae (MF19) to find the Maclaurin's series for y in terms of x up to and including the term in x^4. [2]

....

....

....

....

....

....

....

....

(b) Deduce the value of $\frac{d^4y}{dx^4}$ when $x = 0$. [1]

....

....

....

(c) Use your answer to part **(a)** to find an approximation to $\int_0^{\frac{1}{2}} y \, dx$, giving your answer as a rational fraction in its lowest terms. [2]

....

....

....

....

....

....

....

....

....

....

....

....